① $A = \begin{bmatrix} .8 & .3 \\ .2 & .7 \end{bmatrix}$

$\det(A - \lambda I) = 0$

$$\begin{vmatrix} 0.8-\lambda & 0.3 \\ 0.2 & 0.7-\lambda \end{vmatrix} = 0$$

$(0.8-\lambda)(0.7-\lambda) - 0.06 = 0$

$= \lambda^2 + 0.56 - 1.5\lambda - 0.06 = 0$

$\therefore \lambda^2 - 1.5\lambda + 0.5 = 0$

$\therefore \lambda = 1, \frac{1}{2}$

$A - I$ & $A - \frac{1}{2}I$ not invertible.

eigenvectors x_1, x_2 are in the nullspace of $A - I$ & $A - \frac{1}{2}I$.

$(A - I)x_1 = 0 \quad \therefore Ax_1 = x_1$

$$\begin{bmatrix} 0.8-1 & 0.3 \\ 0.2 & 0.7-1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \quad \therefore$$

$$\begin{bmatrix} -0.2 & 0.3 \end{bmatrix} \begin{bmatrix} x \\ \end{bmatrix} = 0$$

$\begin{bmatrix} 0.2 & -0.3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$

$\therefore \quad -0.2x + 0.3y = 0 \quad \therefore \quad x = \frac{3}{2}y$

$0.2x - 0.3y = 0$

if $y = 2, x = 3 \quad \therefore \vec{x_1} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

normalize $\rightarrow \sqrt{3^2+2^2} = \sqrt{13}$

$$\begin{bmatrix} \frac{3}{\sqrt{13}} \\ \frac{2}{\sqrt{13}} \end{bmatrix} = \begin{bmatrix} 0.832 \\ 0.5547 \end{bmatrix}$$

$$\simeq \begin{bmatrix} 0.6 \\ 0.4 \end{bmatrix}$$

For, $\lambda = \frac{1}{2}$

$$\left(A - \frac{1}{2}I\right) = \begin{bmatrix} 0.3 & 0.3 \\ 0.2 & 0.2 \end{bmatrix} = x + y = 0$$

$\therefore x = -y$

$$\vec{x_2} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix}$$

$Ax_1 = x_1 \quad \therefore \quad A^n x_1 = x_1$

$A^2x_1 = Ax_1$

$A^2x_1 = x_1$

$Ax_2 = \frac{1}{2}x_2 \therefore A^2x_2 = \left(\frac{1}{2}\right)^2 x_2$

$\therefore A^n x = \lambda^n x$

$A^2 = 1^2$ & $\left(\frac{1}{2}\right)^2$

$A^\infty = 1^\infty$ & $0^\infty = 1$ & 0

A now exchange $\rightarrow$ $\lambda = 1, -\frac{1}{2}$

Every A has n-n eigenvals.

(2) $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$

$\lambda_1 = -1 \quad \lambda_2 = 5$

$\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$

$x_1 + 2x_2 = 0$

$x_1 = -2x_2$

$x_2 = -1 \quad , x_1 = 2$

$$\begin{bmatrix} 1-5 & 4 \\ 2 & 3-5 \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$x_2 = 1 \quad , x_1 = 1$

$A+I \longrightarrow$ same eigvectors → $A+I$ singular

eigval $= -1+1 = 0$

& $5+1 = 6$

You can think of A as doing something geometric: stretching, rotating, squashing certain directions.

Adding I to it is like saying, "Okay, now every vector also gets pulled a little bit in its own direction." This affects the amount of stretching (the eigenvalue), but not which directions get stretched (the eigenvectors).

(3) A & A^{-1} reciprocal e.val

→ A does, A^{-1} undoes

Direction same

(5) For triangular mat, λ = pivots.

(6) Eignvald AB = Eval of BA

(8) Evec given, $Ax = \lambda x$ ∴ $x =$

Eval :: $\text{solve}(A - \lambda I) = 0$

(9) a) $Ax = \lambda x$

$A^2x = \lambda Ax$

$A^2x = \lambda\lambda x$

$A^2x = \lambda^2 x$

b) $Ax = \lambda x$

$AA^{-1}x = A^{-1}\lambda x$

$x = A^{-1}\lambda x$

$\frac{x}{\lambda} = A^{-1}x$

$\therefore A^{-1}x = \frac{1}{\lambda}x$

c) $Ax = \lambda x$

$Ax + Ix = \lambda x + Ix$

$(A+I)x = \lambda x + x$

$= (\lambda + 1)x$

(10) $A = \begin{bmatrix} .6 & .2 \\ .4 & .8 \end{bmatrix}$ $(A - \lambda I)$ $= \lambda^2 - 1.4\lambda + 0.4$

$= 0$

$\therefore \lambda =$ 1, 0.4

$x_1 = (1, 2)$; $x_2 = (1, -1)$

A^∞ : $\lambda = 1^\infty, 0.4^\infty = 1, 0$

Evec same -

A^{100} is very near A^∞ same evec close eval.

(1) $A_{2\times 2}$ Mat $\rightarrow$ $\lambda_1 \neq \lambda_2$

So $v_1 \neq v_2$ (linearly independent Evectors)

$A v_1 = \lambda_1 v_1$ & $A \cdot v_2 = \lambda_2 v_2$

Now $(A-\lambda_1 I)x = Ax - \lambda_1 x$

$= Av_1 - \lambda_1 v_1$

$= \lambda_1 v_1 - \lambda_1 v_1$

$= 0$

So v_1 is in Nullspace of $A - \lambda_1 I$

Let, A be a diagonal Mat. eval $\lambda_1 \lambda_2$

$$\therefore A - \lambda_1 I = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} - \lambda_1 I$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & \lambda_2 - \lambda_1 \end{bmatrix}$$

$$(A - \lambda_1 I) v_1 = 0$$

$$(A - \lambda_1 I) v_2 = A v_2 - \lambda_1 v_2$$

$$= \lambda_2 v_2 - \lambda_1 v_2$$

$$= (\lambda_2 - \lambda_1) v_2$$

So the Mat $(A - \lambda_1 I)$ map v_1 to 0 & v_2 to scalar multiple of itself(v_2)

So the image of $A - \lambda_1 I$ is the span of v_2.

Given a 2x2 matrix A with distinct eigenvalues λ_1 and λ_2, and corresponding eigenvectors v_1 and v_2, we want to understand why the columns of $(A - \lambda_1 \cdot I)$ are multiples of the eigenvector v_2.

- Eigenvalue Subtraction

Subtracting $\lambda_1 \cdot I$ from A gives the matrix $(A - \lambda_1 \cdot I)$. For eigenvector v_1 of λ_1:

$$(A - \lambda_1 \cdot I) \cdot v_1 = 0$$

This means v_1 is in the null space of $(A - \lambda_1 \cdot I)$.

- **Action on Eigenvector v_2**

For eigenvector v_2 of λ_2:

$$(A - \lambda_1 \cdot I) \cdot v_2 = (\lambda_2 - \lambda_1) \cdot v_2$$

So v_2 is just scaled and stays in the same direction.

- **Column Space**

Since v_1 is in the null space, and the null space is 1D, the rank of $(A - \lambda_1 \cdot I)$ is 1.

That means the column space is also 1D.

Because v_2 is not in the null space, the column space must be in the direction of v_2.

- **Columns as Multiples**

The columns of $(A - \lambda_1 \cdot I)$ are what you get by applying the matrix to basis vectors.

Since the column space is along v_2, each column is a multiple of v_2.

Conclusion

The matrix $(A - \lambda_1 \cdot I)$ has rank 1, and its columns span the eigenspace of λ_2.

$\Rightarrow$ All columns are multiples of v_2.

Further by shara:

Here's a Samsung Notes–friendly plain-text version you can copy-paste:

Given a 2×2 matrix A with distinct eigenvalues λ_1 and λ_2, and eigenvectors v_1 and v_2, we show why the columns of (A –

$\lambda_2 \cdot I$) are multiples of v_1.

- **Null space**

$(A - \lambda_2 \cdot I) \cdot v_2 = 0$

$\Rightarrow$ v_2 lies in the null space of $(A - \lambda_2 \cdot I)$.

- **Action on v_1**

$(A - \lambda_2 \cdot I) \cdot v_1$

$= A \cdot v_1 - \lambda_2 \cdot v_1$

$= \lambda_1 \cdot v_1 - \lambda_2 \cdot v_1$

$= (\lambda_1 - \lambda_2) \cdot v_1$

$\Rightarrow$ v_1 is sent to a nonzero multiple of itself.

- **Rank = 1**

Null space is 1-D $\Rightarrow$ rank = 1 $\Rightarrow$ column space is 1-D.

- **Columns span v_1**

Since $v_1 \notin$ null space, it spans the column space.

$\Rightarrow$ each column of $(A - \lambda_2 \cdot I)$ is a multiple of v_1.

Conclusion

$(A - \lambda_2 \cdot I)$ has rank 1 and its columns span the eigenspace of λ_1.

(12)

$$P = \begin{bmatrix} .2 & .4 & 0 \\ .4 & .8 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Check after 16!

$$P^2 = \begin{bmatrix} .2 & .4 & 0 \\ .4 & .8 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

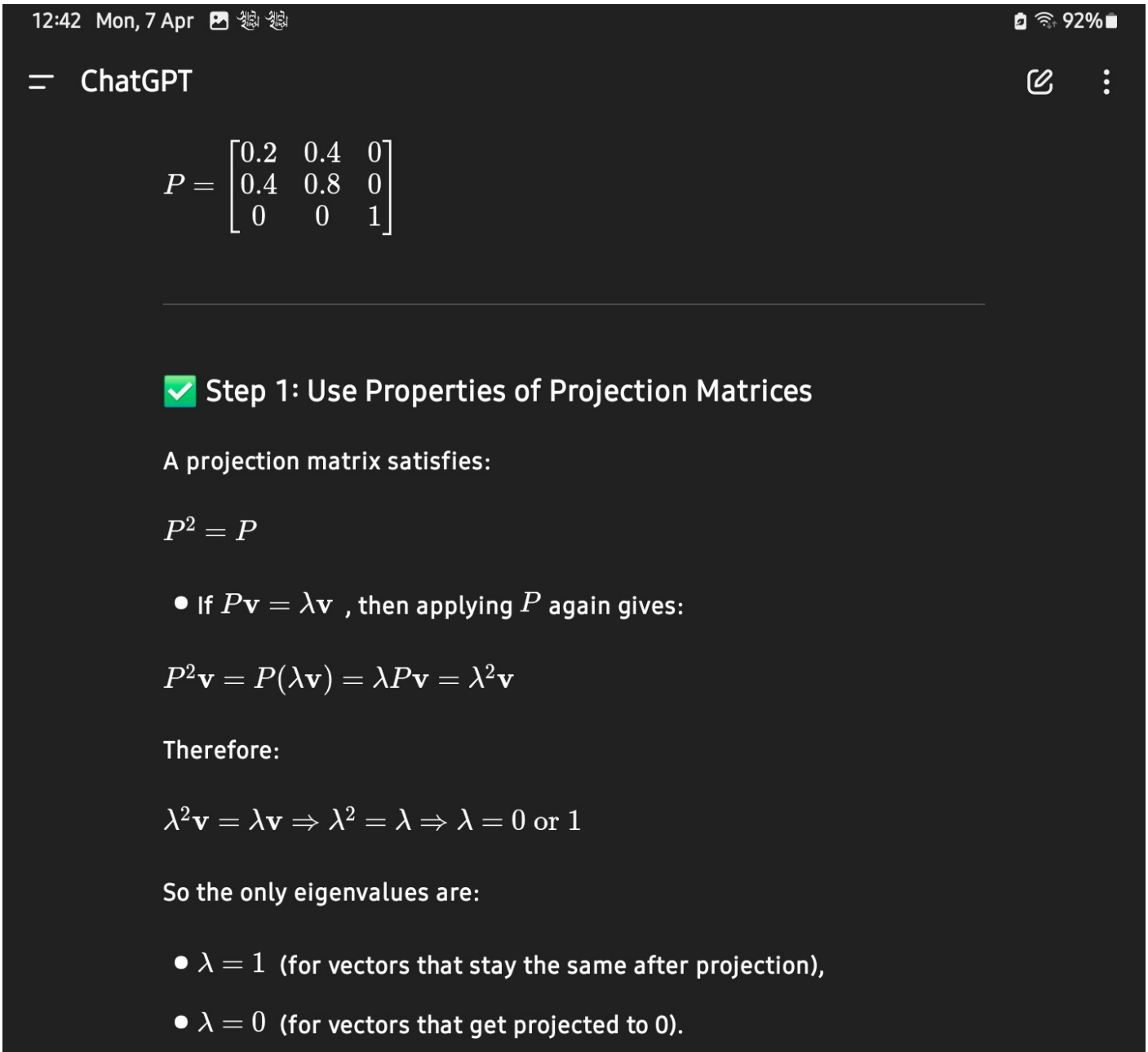
trace = $.2 + .8 + 1 = 2$ = rank for Proj Mat.

2 ones & 1 zero.

1, 1, 0

$\lambda = 1$ (with multiplicity 2) $\lambda = 0$

The projection matrix P has λ = 1, 0, 1 with eigenvectors(1, 2, 0), (2, −1, 0), (0, 0, 1).
Add the first and last vectors: (1, 2, 1) also has λ = 1. The whole column space of P
contains eigenvectors with λ = 1 ! Note P
2 = P leads to λ
2 = λ so λ = 0 or 1.



✅ Step 2: Find Eigenvectors for $\lambda = 1$

These vectors lie in the image of P (column space). To find them:

- Let's observe the first two rows and columns of P:

$$\begin{vmatrix} 0.2 & 0.4 \\ 0.4 & 0.8 \end{vmatrix}$$

We can factor this part of P as a rank-one projection, meaning it projects onto a line.

Try the vector:

$$\begin{bmatrix} 1 \end{bmatrix}$$

Ask anything

12:42 Mon, 7 Apr 92%

ChatGPT

Try the vector:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$P\mathbf{v}_1 = \begin{bmatrix} 0.2 & 0.4 & 0 \\ 0.4 & 0.8 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.2(1) + 0.4(2) \\ 0.4(1) + 0.8(2) \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \mathbf{v}_1$$

✅ So, $\mathbf{v}_1$ is an eigenvector with $\lambda = 1$

Similarly, try:

$$\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow P\mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \mathbf{v}_2$$

✅ So, $\mathbf{v}_2$ is also an eigenvector with $\lambda = 1$

✅ Step 3: Find Eigenvector for $\lambda = 0$

These lie in the null space of P, meaning:

$$P\mathbf{v} = 0$$

We can solve the system:

$$\begin{bmatrix} 0.2 & 0.4 & 0 \\ 0.4 & 0.8 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} 0.2x + 0.4y = 0 \\ 0.4x + 0.8y = 0 \\ z = 0 \end{cases}$$

Solving:

- From first equation: $0.2x + 0.4y = 0 \Rightarrow x = -2y$
- From last: $z = 0$

So a solution is:

$$\mathbf{v}_2 = \begin{bmatrix} -2 \\ 1 \\ \end{bmatrix} \rightarrow P\mathbf{v}_2 = \mathbf{0} \rightarrow \lambda = 0$$

Ask anything

12:42 Mon, 7 Apr 92%

ChatGPT

✅ Step 4: Get an Eigenvector with No Zero Components

We want a vector that:

- Has no zero entries.
- Is still an eigenvector (preferably with $\lambda = 1$).

We combine $\mathbf{v}_1 = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix}$ and
$\mathbf{v}_2 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$

Add:

$$\mathbf{v}_{\text{non-zero}} = \mathbf{v}_1 + \mathbf{v}_2 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Check:

$$P \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.2(1) + 0.4(2) \\ 0.4(1) + 0.8(2) \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \text{still the same vector}$$

✅ So it's an eigenvector with eigenvalue $\lambda = 1$, and no zero entries.

✅ Final Summary

- Three eigenvectors:
  a. $[1\ 2\ 0]\ -\lambda = 1$
  b. $[0\ 0\ 1]\ -\lambda = 1$
  c. $[-2\ 1\ 0]\ -\lambda = 0$
- Eigenvector with no zero components:

$$\begin{bmatrix}1\\2\\1\end{bmatrix} \quad (\lambda = 1)$$

Ask anything

(13) a) $P = uu^T$, $u^Tu = 1$

$\therefore Pu = (uu^T)u = u(u^Tu) = u(1) = u$

$\therefore \lambda = 1$

b) $v \perp u$, $u^Tv = 0$

$Pv = (uu^T)v = u(u^Tv) = 0$

So, $Pv = 0$

↓

v a eigvec with eval $= 0$. $\lambda = 0$

c) 3 ind. e.vec of P with $\lambda = 0$ $u(1/6, 1/6, 3/6, 4/6)$

given, $P \rightarrow$ rank 1 proj Mat. in $\mathbb{R}^4$

its Nullspace is 3d. $(\lambda = 0)$

$$u = \left(\frac{1}{6}, \frac{1}{6}, \frac{3}{6}, \frac{5}{6}\right) \rightarrow \text{unit vect.}$$

$$= \frac{1}{6}(1, 1, 3, 5)$$

$$u^T x = 0$$

$$\therefore \frac{1}{6}(x_1 + x_2 + 3x_3 + 5x_4) = 0$$

$$x_1 + x_2 + 3x_3 + 5x_4 = 0$$

$$x_1 = -x_2 - 3x_3 - 5x_4$$

taking (x_2, x_3, x_4) 1 at a time as free vector :

let $(x_2, x_3, x_4) = (1, 0, 0)$

$$x_1 = -1 \quad , \quad x = (-1, 1, 0, 0)$$

let $(x_2, x_3, x_4) = (0, 1, 0)$

$x_1 = -3$, $x(-3, 0, 1, 0)$

Let $(x_2, x_3, x_4) = (0, 0, 1)$

$x_1 = -5$, $x(-5, 0, 0, 1)$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \rightarrow \text{Linearly indep.}$$

(14) $Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ $\det(Q - \lambda I) = 0$

$$\begin{bmatrix} \cos\theta - \lambda & -\sin\theta \\ \sin\theta & \cos\theta - \lambda \end{bmatrix}$$

$\lambda^2 - 2\lambda\cos\theta + 1 = 0$

$\lambda^2 - \text{trace}\,\lambda + \det$

$\dfrac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2}$

$\cos\theta \pm \sqrt{-\sin^2\theta} = \cos\theta \pm i\sin\theta$

$= e^{i\theta}$ & $e^{-i\theta}$

$$\begin{bmatrix} i\sin\theta & -\sin\theta \\ \sin\theta & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$i\sin\theta x_1 - \sin\theta x_2 = 0$$

$$\therefore \quad ix_1 = x_2$$

$$x_1 = 1, \quad \therefore \quad x = (1, i)$$

for $\lambda = \cos\theta + i\sin\theta$, $x_1 = -ix_2$

$$\begin{bmatrix} -i\sin\theta & -\sin\theta \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$-ix_1 = x_2$$

$$-i(1) = x_2$$

$$\therefore \quad x = (1, -i)$$

(15) $P_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

3 cycle $(1 \to 2 \to 3 \to 1)$: char polynomial:

$$\det(P_1 - \lambda I) = -(\lambda^3 - 1) = 0$$

$$\lambda^3 = 1$$

$\therefore \lambda = 1, \quad e^{2\pi i/3} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$)

$$e^{4\pi i/3} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$$

$\therefore$ beyond $\lambda = 1$ two more one,

$$-\frac{1}{2} \pm i\frac{\sqrt{3}}{2}$$

$$P_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & -\lambda \end{bmatrix} = +(1-\lambda)(\lambda^2 - 1)$$

$$= (1-\lambda)(\lambda+1)(\lambda-1)$$

$$=$$

$1 \to 3 \to 1$

Laplace cofactor expansion

Laplace Cofactor Expansion

$$P_1 = \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{pmatrix}$$

$$P_1 - \lambda I = \begin{pmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & 0 & -\lambda \end{pmatrix}$$

$$-\lambda(\lambda^2) - 1(0-1)$$

$$= -\lambda^3 + 1 = -(\lambda^3 - 1)$$

$$\det(P_2 - \lambda I) = (\lambda+1)(\lambda-1)(1-\lambda)$$

$$\therefore \lambda = 1, -1$$

$$1 - \lambda = 0 \therefore \lambda = 1$$

2 more besides $\lambda = 1$ once,

$$\lambda = 1, -1$$

16 For $A_{n\times n}$ Mat, evals $\lambda_1 \cdots \lambda_n$

$$\det(A-\lambda I) = (\lambda_1-\lambda)(\lambda_2-\lambda)\cdots(\lambda_n-\lambda)$$

$\lambda = 0$ give

$$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n$$

$$= \prod_{i=1}^{n} \lambda_i$$

Example 1

$$\det(A) = 1 \times \frac{1}{2} = \frac{1}{2}$$

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

$$= 0.56 - 0.6 = \frac{1}{2}$$ Alhamdulillah

$$\lambda^2 - \text{trace}\lambda + \det$$

18:12 Mon, 7 Apr • 91%
ChatGPT

First, compute the characteristic polynomial of

$$P = \begin{pmatrix} 0.2 & 0.4 & 0 \\ 0.4 & 0.8 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

P-\lambda I =\begin{pmatrix} 0.2-\lambda & 0.4 & 0\ 0.4 & 0.8-\lambda& 0\ 0 & 0 & 1-\lambda \end{pmatrix},] so

$$\det(P-\lambda I) = (1-\lambda)\,\det\begin{pmatrix} 0.2-\lambda & 0.4 \\ 0.4 & 0.8-\lambda \end{pmatrix} = (1-\lambda)\Big[(0.2-\lambda)(0.8-\lambda) - 0.4^2\Big].$$

$$(0.2-\lambda)(0.8-\lambda) - 0.16 = \lambda^2 - (0.2+0.8)\lambda + (0.16-0.16) = \lambda^2 - \lambda.$$

$$\det(P-\lambda I) = (1-\lambda)\,(\lambda^2-\lambda) = (1-\lambda)\,\lambda\,(\lambda-1) = -\lambda\,(1-\lambda)^2.$$

$$\boxed{\lambda = 0 \quad (\text{mult. } 1), \quad \lambda = 1 \quad (\text{mult. } 2).}$$

---

1. Eigenvector for $\lambda = 0$

Solve $P, v = 0$:

$$\begin{cases} 0.2\,v_1 + 0.4\,v_2 = 0, \\ 0.4\,v_1 + 0.8\,v_2 = 0, \\ v_3 = 0. \end{cases}$$

$v^{(0)} = (-2,\, 1,\, 0)^T$

---

2. Eigenvectors for $\lambda = 1$

Solve $(P-I), v = 0$:

$$\begin{pmatrix} -0.8 & 0.4 & 0 \\ 0.4 & -0.2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 0 \implies \begin{cases} -0.8\,v_1 + 0.4\,v_2 = 0 \;\Rightarrow\; v_2 = 2v_1, \\ v_3 \text{ is free.} \end{cases}$$

$v = \big(v_1,\, 2v_1,\, v_3\big) = v_1\,(1,2,0)^T$ $_3\,(0,0,1)^T.$

Ask anything

18:12 Mon, 7 Apr • 91%

Share a link to chat?
This creates a copy that others can chat with

$$\begin{cases} 0.2\,v_1 + 0.4\,v_2 = 0, \\ 0.4\,v_1 + 0.8\,v_2 = 0, \\ v_3 = 0. \end{cases}$$

$v^{(0)} = (-2,\,1,\,0)^T$

---

2. Eigenvectors for $\lambda = 1$

Solve $(P - I), v = 0$:

$$\begin{pmatrix} -0.8 & 0.4 & 0 \\ 0.4 & -0.2 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = 0 \implies \begin{cases} -0.8\,v_1 + 0.4\,v_2 = 0 \;\Rightarrow\; v_2 = 2v_1, \\ v_3 \text{ is free.} \end{cases}$$

$v = \big(v_1,\,2v_1,\,v_3\big) = v_1\,(1,2,0)^T \;+\; v_3\,(0,0,1)^T.$

$v_1^{(1)} = (1,2,0)^T, \qquad v_2^{(1)} = (0,0,1)^T.$

---

3. “No-zero-entry” eigenvector for $\lambda = 1$

Any nontrivial linear combination $\alpha, v_1^{(1)} + \beta, v_2^{(1)}$ is still an eigenvector with $\lambda = 1$. To get no zero components, pick $\alpha = \beta = 1$:

$v^{(1)} = (1,2,0) + (0,0,1) = (1,\,2,\,1)^T.$

$P\,(1,2,1)^T = (1,2,1)^T,$

Summary of three eigenvectors

- $\lambda = 0 :; (-2,1,0)^T$.
- $\lambda = 1 :; (1,2,0)^T,; (0,0,1)^T$.
- “No-zero-entry” $\lambda = 1$ vector: $(1,2,1)^T$.

Ask anything

(17) $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\det(A - \lambda I) = \lambda^2 - (a+d)\lambda + ad - bc$$

$$\lambda_1 + \lambda_2 = a + d = +\text{tr}(A)$$

$$\lambda_1 \lambda_2 = ad - bc = \det(A)$$

(18) $\begin{bmatrix} 3 & 2 \\ -1 & 6 \end{bmatrix}$ $\begin{bmatrix} 4 & 0 \\ 0 & 5 \end{bmatrix}$, $\begin{bmatrix} 4 & 6 \\ 0 & 5 \end{bmatrix}$

(19) A 3×3 Mat with e.val = 0, 1, 2

(i) rank of B = 2 (2 non zero e.val

(ii) Det of B^TB

Fact:

$$\det B^TB = (\det B)^2$$

$$= (0 \cdot 1 \cdot 2)^2 = 0^2 = 0$$

(iii) eig vals of $B^TB = 0, 1, 2 = 0, 1, 2$

(iv) " " $(B^2+I)^{-1} = \frac{1}{0+1}, \frac{1}{1+1}, \frac{1}{4+1}$

$= 1, \frac{1}{2}, \frac{1}{5}$

$$\begin{bmatrix} 2 & 1 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 5 & 7 \\ 7 & 10 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 & 2 \\ 6 & 2 & 4 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \end{bmatrix}$$

(20) Companion Matrix: monic polynomial.

(poly with leading coeff 1)

$$P(\lambda) = \lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

$$C = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \vdots & \ddots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

For Mat with $\lambda = 1, 2, 3$

$(\lambda-1)(\lambda-2)(\lambda-3) = \lambda^3 - 6\lambda^2 + 11\lambda - 6$

$= \lambda^3 + (-6)\lambda^2 + 11\lambda + (-6)$

$-a_0, -a_1, -a_2 = 6, -11, 6$

$$C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{bmatrix}$$

21

19:27 Mon, 7 Apr • 87%

ChatGPT

🧠 Think of It Like This:

- A and A^t can have different directions of stretching (eigenvectors), even though the amount of stretching (eigenvalues) is the same.
- They preserve the same volumes, but not necessarily the same directions.

📌 Example

Take this matrix:

$$A = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix}, \quad A^T = \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}$$

They both have eigenvalues 0 and 1, but:

- A's eigenvectors are like $[1\ 1]$
- A^T's eigenvectors are like $[1\ 0]$

See? Same stretch factor, but different direction.

💡 Summary

Property	A vs. A^T
Eigenvalues	Always the same
Eigenvectors	Usually different

Because the transpose flips rows and columns, it affects how vectors are multiplied, which changes the eigenvectors, but not the eigenvalues.

Ask anything

22

🔷 1. Matrix M with eigenvalues on the diagonal

You wrote a matrix like this:

$$M = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0.4 & 0 \\ 0.7 & 0.6 & 1 \end{bmatrix}$$

This matrix is upper triangular, and here's the rule:

> ✅ The eigenvalues of a triangular matrix (upper or lower) are just its diagonal entries.

So yes, this matrix has eigenvalues:

$$\lambda_1 = 1, \quad \lambda_2 = 0.4, \quad \lambda_3 = 1.0$$

🔷 2. Why $M^T\mathbf{v} = \mathbf{v}$ when $\mathbf{v} = [1\ 1\ 1]$

If all rows of M^T add to 1, then M^T is a stochastic matrix (like a Markov matrix — each column sums to 1, or each row in the transposed case). This means:

$$M^T \cdot \begin{bmatrix}1\\1\\1\end{bmatrix} = \begin{bmatrix}1\\1\\1\end{bmatrix}$$

✅ So $\mathbf{v} = [1\ 1\ 1]$ is an eigenvector of M^T with eigenvalue $\lambda = 1$.

🔷 3. The Challenge: A 3×3 Singular Markov Matrix with Trace = 2

You're asked to create or understand a 3×3 Markov matrix (each row adds to 1), that's also:

- Singular (determinant = 0 → at [illegible] one eigenvalue = 0)
- Has eigenvalues:

Ask anything

19:39 Mon, 7 Apr • 87%
ChatGPT

🔷 3. The Challenge: A 3×3 Singular Markov Matrix with Trace = 2

You're asked to create or understand a 3×3 Markov matrix (each row adds to 1), that's also:

- Singular (determinant = 0 → at least one eigenvalue = 0)
- Has eigenvalues:

$$\lambda_1 = 1, \quad \lambda_2 = 0, \quad \lambda_3 = -\tfrac{1}{2}$$

Let's verify this makes sense.

🔍 Trace = Sum of Eigenvalues

$$\text{Trace} = \lambda_1 + \lambda_2 + \lambda_3 = 1 + 0 + (-\tfrac{1}{2}) = \tfrac{1}{2}$$

But your prompt says the trace is 2... 🤔 That doesn't match. Maybe a typo?

But okay — let's understand why a Markov matrix can have negative eigenvalues.

🔷 Markov Matrix Properties

- Real eigenvalue = 1 is always present (represents the steady state).
- Other eigenvalues can be < 1, even negative, but their absolute value must be ≤ 1 (from the Perron–Frobenius theorem).
- If one eigenvalue is 0, the matrix is singular (non-invertible).
- If all rows sum to 1, then $A \cdot \mathbf{1} = \mathbf{1}$, so 1 is an eigenvalue.

So yes, a Markov matrix can have eigenvalues $1, 0, -\frac{1}{2}$.

Ask anything

23

19:56 Mon, 7 Apr • 86%

ChatGPT

You are looking for three 2×2 matrices A that satisfy the following conditions:

1. $A^2 = 0$ (the square of the matrix is the zero matrix).
2. The trace of A is 0.
3. The determinant of A is 0.
4. $A \neq 0$ (i.e., A is not the zero matrix).

Additionally, we will be using the Cayley-Hamilton theorem, which

states that every square matrix satisfies its own characteristic equation.

Matrix 1:

$$A_1 = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix}$$

1. Check A_1^2:

$$A_1^2 = \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 0 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$$

1. Check the trace of A_1: The trace is the sum of the diagonal elements:

$$\text{trace}(A_1) = 0 + 0 = 0$$

1. Check the determinant of A_1: The determinant of a 2×2 matrix is given by $\det(A_1) = ad - bc$, where the matrix is $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A_1) = 0 \times 0 - 0 \times 1 = 0$$

Thus, A_1 satisfies all the conditions.

Matrix 2:

$$A_2 = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}$$

1. Check A_2^2:

$$A_2^2 = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} \cdot \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix}$$

Ask anything

24

$$A = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 2 & 1 & 2 \end{bmatrix}$$

rank $= 1$

trace $= 6$

$\lambda = 0, 0, 6$

$$A = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 2 \\ 4 & 2 & 4 \\ 2 & 1 & 2 \end{bmatrix}$$

$\downarrow$ $\quad$ $\downarrow$

u $\quad$ V^T

$$A = uv^T \qquad u = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \quad v = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$\lambda = 6$:

$$Ax_3 = uv^T x_3$$
$$= u(v^T x_3)$$

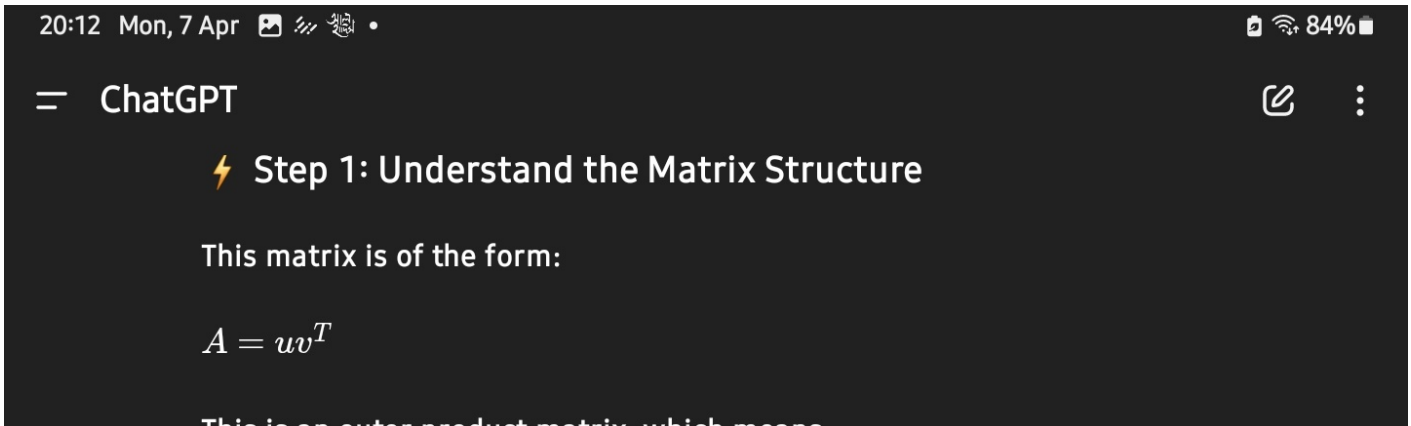
↳ multiple of u

e.vec $\Rightarrow$ $x_3 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = u$

$\lambda = 0$, if & only if $v^T x = 0$

$$v^T x = 2x_1 + x_2 + 2x_3 = 0$$
$$0 \quad -2 \quad 1$$

$$x_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$



This is an outer product matrix, which means:

- It has rank 1
- It has only one nonzero eigenvalue (the rest are 0)

Let's find those eigenvectors!

🧠 Step 2: Eigenvalue $\lambda = 6$

This eigenvalue corresponds to the image of the matrix. Since:

$$A = uv^T \Rightarrow Ax = u(v^T x)$$

If you try $x = u$, then:

$$Au = u(v^T u)$$

Let's compute $v^T u$:

$$v^T u = [2\ 1\ 2] \cdot [1\ 2\ 1] = 2(1) + 1(2) + 2(1) = 2 + 2 + 2 = 6 \Rightarrow Au = 6u$$

So:

$$x_3 = u = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \text{is an eigenvector for} \quad \lambda = 6$$

🧠 Step 3: Eigenvalue $\lambda = 0$

From the structure $Ax = u(v^T x)$ you can see:

- $Ax = 0$ if and only if $v^T x = 0$

Ask anything

20:12 Mon, 7 Apr • 84%

ChatGPT

We're trying to find vectors x such that:

$$v^T x = 0 \quad \Rightarrow \quad 2x_1 + 1x_2 + 2x_3 = 0$$

We want to find two independent vectors in this plane (since the null space of A is 2D), and any two linearly independent solutions to this

equation will work.

✅ How I Picked Those Vectors

Let's solve:

$$2x_1 + x_2 + 2x_3 = 0$$

We can assign free values to two of the variables and solve for the third. That's the standard way to find vectors in a plane:

🔹 First try:

Set $x_1 = 0$, $x_3 = 1$

Then the equation becomes:

$$2(0) + x_2 + 2(1) = 0 \Rightarrow x_2 = -2$$

So we get:

$$x_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

🔹 Second try:

Set $x_1 = 1$, $x_3 = 0$

Equation becomes

+ Ask anything

20:14 Mon, 7 Apr 84%
ChatGPT

🔹 Second try:

Set $x_1 = 1$, $x_3 = 0$

Equation becomes:

$$2(1) + x_2 + 0 = 0 \Rightarrow x_2 = -2$$

So we get:

$$x_2 = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$$

✅ Why These Are Orthogonal to v

Let's verify with a dot product:

$$v^T x_1 = [2\ 1\ 2] \cdot [0\ \ -2\ 1] = 0 - 2 + 2 = 0$$
$$v^T x_2 = [2\ 1\ 2] \cdot [1\ \ -2\ 0] = 2 - 2 + 0 = 0$$

✅ So both are orthogonal to v

🔁 Summary

We found eigenvectors for $\lambda = 0$ by:

1. Writing the orthogonality condition:
   $v^T x = 2x_1 + x_2 + 2x_3 = 0$
2. Picking values for two variables (free choice), solving for the third.
3. Verifying dot product = 0 (orthogonal).

That's it! Let me know if you want ... visualize the null space or the projection on the direction of u.

Ask anything

20:13 Mon, 7 Apr • 84%
ChatGPT

Given Orthogonality Condition:

We want to find all vectors $x = [x_1\ x_2\ x_3]$ such that:

$$v^T x = 2x_1 + x_2 + 2x_3 = 0$$

This is a single linear equation, so the solution space is a plane. If you treat x_1 and x_2 as free variables, you can solve for x_3.

✅ Step-by-step:

We solve:

$$2x_1 + x_2 + 2x_3 = 0 \quad \Rightarrow \quad x_3 = -\tfrac{1}{2}(2x_1 + x_2)$$

So any vector of the form:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ -\frac{1}{2}(2x_1 + x_2) \end{bmatrix}$$

🔧 Example: Choose specific values

Case 1: $x_1 = 1$, $x_2 = 0$

$$x_3 = -\tfrac{1}{2}(2 \cdot 1 + 0) = -1 \Rightarrow x = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

Case 2: $x_1 = 0$, $x_2 = 1$

$$x_3 = -\tfrac{1}{2}(0 + 1) = -\tfrac{1}{2} \Rightarrow x = \begin{bmatrix} 0 \\ 1 \\ -\frac{1}{2} \end{bmatrix}$$

These are both valid eigenvectors with $\lambda = 0$, and since they are linearly independent, they span t[illegible]l space.

Ask anything

26

20:21 Mon, 7 Apr 84%

ChatGPT

- B is a 2×2 matrix (with eigenvalues 1 and 2)
- C is a 2×2 matrix (but doesn't affect eigenvalues)
- D is a 2×2 matrix (with eigenvalues 5 and 7)

So overall:

$$A = \begin{vmatrix} B & C \\ 0 & D \end{vmatrix} \quad (4\tilde{A}\blacksquare 4 \text{ matrix})$$

💡 Key Idea: Block Triangular Matrices

If a matrix is block upper-triangular like:

$$\begin{vmatrix} B & C \\ 0 & D \end{vmatrix}$$

Then its eigenvalues are just the eigenvalues of the diagonal blocks, i.e., the eigenvalues of B and D.

✅ This is because the characteristic polynomial factors like:

$$\det(\lambda I - A) = \det(\lambda I - B) \cdot \det(\lambda I - D)$$

🧠 So What Are the Eigenvalues?

From the diagonal blocks:

- B has eigenvalues: 1, 2
- D has eigenvalues: 5, 7

Then:

$$\boxed{\text{Eigenvalues of } A: \quad \lambda = 1, 2, 5, 7}$$

+ Ask anything

27 $A =$ 

→ rank 1

eigval = 0, 0, 0, 4 ↓

trace

20:29 Mon, 7 Apr 83%

ChatGPT

🧮 Matrix A:

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

🔹 Rank of A:

Every row is identical (or a scalar multiple of another), so the row space has dimension 1.

✅ So,

$$\boxed{\text{rank}(A) = 1}$$

🔹 Eigenvalues of A:

Let's reason this way:

- A has rank 1, so only one non-zero eigenvalue, the rest must be 0.
- A is symmetric, so it's diagonalizable, and its eigenvalues are real.
- Let's find a non-zero eigenvalue:

Notice:

$$A \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 4 \\ 4 \end{bmatrix} = 4 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

✅ So eigenvector $\vec{v} = [1\ 1\ 1\ 1]^T$ has eigenvalue $\lambda = 4$

The trace of A is 4, and since the sum of eigenvalues equals the trace:

$$\text{Eigenvalues of } A: \boxed{4,\ 0,\ 0,\ 0}$$

Ask anything

20:30 Mon, 7 Apr 83%

ChatGPT

$$\begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

This matrix is symmetric, so again all eigenvalues are real.

🔹 Rank of C:

Rows 1 and 3 are the same.
Rows 2 and 4 are the same.

So the row space is 2D, meaning:

$$\boxed{\text{rank}(C) = 2} \quad \Rightarrow \quad 2 \text{ zero eigenvalues}$$

🔹 Non-zero eigenvalues?

Try:

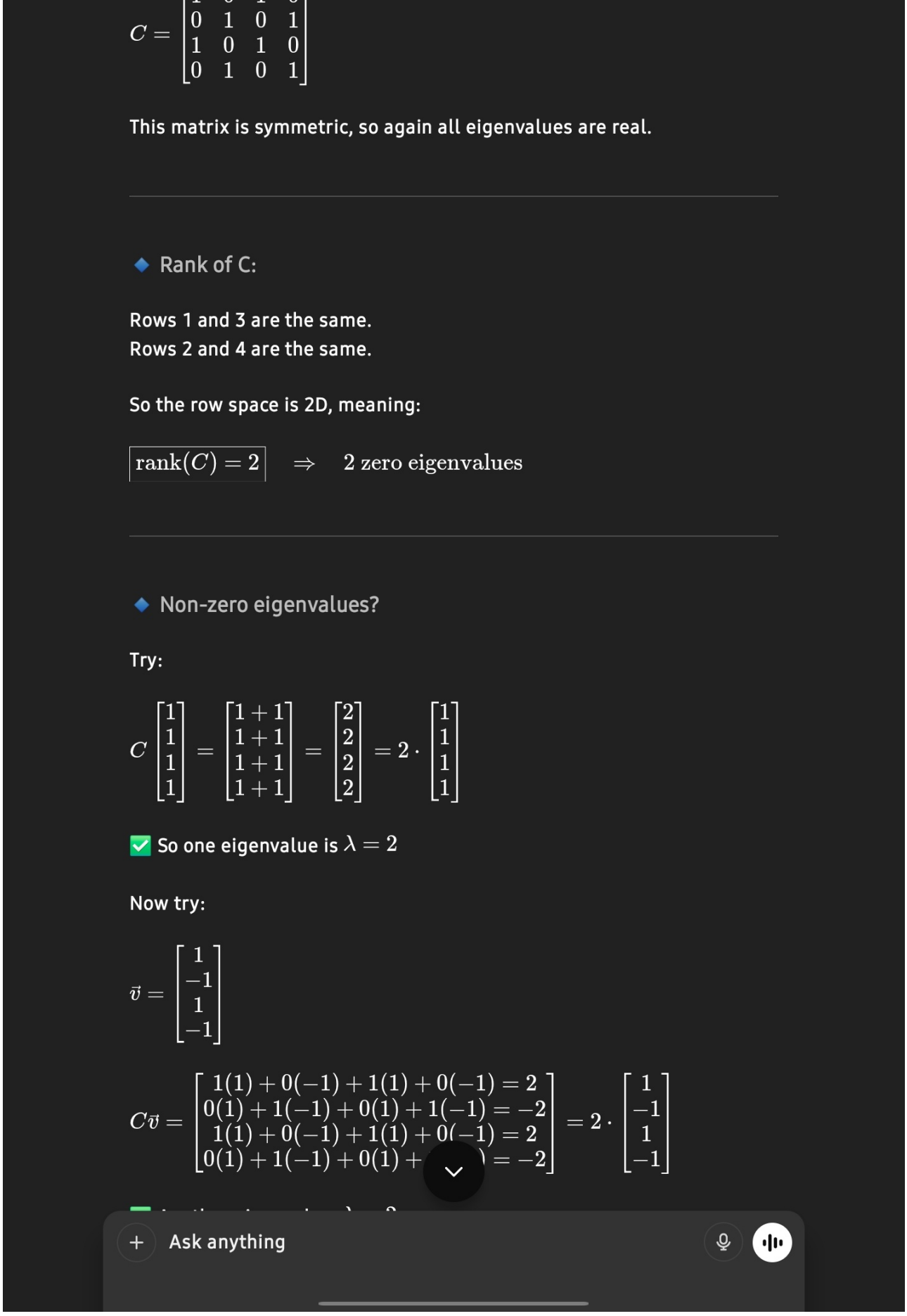
$$C \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1 \\ 1+1 \\ 1+1 \\ 1+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \\ 2 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

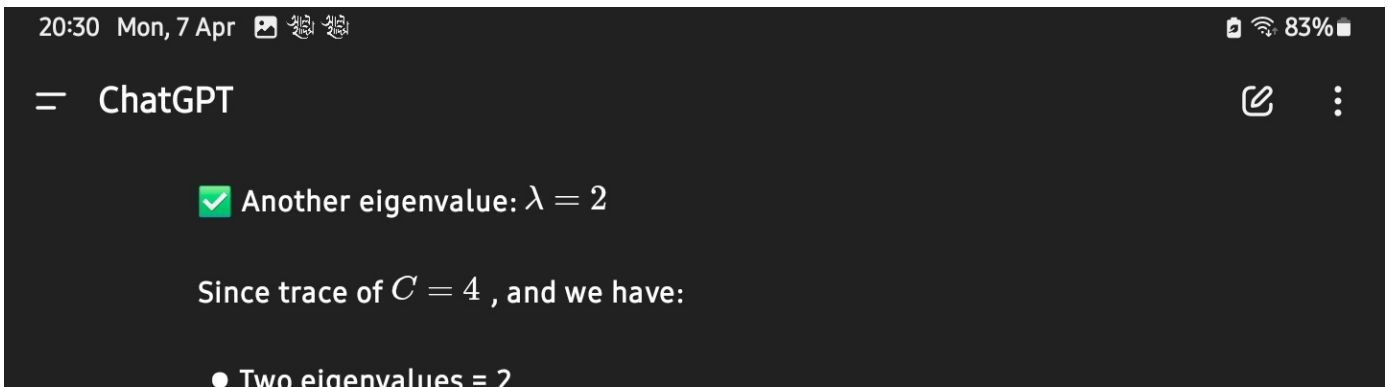
✅ So one eigenvalue is $\lambda = 2$

Now try:

$$\vec{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

$$C\vec{v} = \begin{bmatrix} 1(1) + 0(-1) + 1(1) + 0(-1) = 2 \\ 0(1) + 1(-1) + 0(1) + 1(-1) = -2 \\ 1(1) + 0(-1) + 1(1) + 0(-1) = 2 \\ 0(1) + 1(-1) + 0(1) + \quad) = -2 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$





- Remaining two must be zero

$$\boxed{\text{Eigenvalues of } C:\ 2,\ 2,\ 0,\ 0}$$

🔁 Summary

Matrix	Rank	Eigenvalues	Non-zero Eigenvectors
A	1	4, 0, 0, 0	$[1\ 1\ 1\ 1]^T$
C	2	2, 2, 0, 0	$[1\ 1\ 1\ 1]^T$, $[1\ -1\ 1\ -1]^T$

Let me know if you'd like to see eigenvectors for the zero eigenvalues or a full diagonalization!

eigenvectors too

Awesome — let's find all eigenvectors of matrices A and C, including for the zero eigenvalues. We'll walk through this step-by-step.

🧮 Matrix A

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

We already know:

- $\lambda_1 = 4$, eigenvector: $\vec{v}_1 = [1\ 1\ 1\ 1]$

Ask anything

20:31 Mon, 7 Apr 83%

ChatGPT

- $\lambda = 0$ has multiplicity 3 — we want to find 3 independent eigenvectors for it

---

🔍 For $\lambda = 0$: Solve $A\vec{x} = 0$

Since all rows of A are the same, the null space consists of all vectors orthogonal to $\vec{1} = [1\ 1\ 1\ 1]$.

So find vectors x such that:

$$x_1 + x_2 + x_3 + x_4 = 0$$

We can construct a basis of 3 eigenvectors like:

- $\vec{v}_2 = [1\ -1\ 0\ 0]$
- $\vec{v}_3 = [0\ 1\ -1\ 0]$
- $\vec{v}_4 = [0\ 0\ 1\ -1]$

Each of these satisfies $A\vec{v}_i = 0$, so:

$$\text{Eigenvectors of A:} \quad \lambda = 4: \quad \vec{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\lambda = 0: \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \ \vec{v}_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \ \vec{v}_4 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

🧮 Matrix C

$$C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

We already know:

+ Ask anything

20:31 Mon, 7 Apr 83%

ChatGPT

- Eigenvalues: $\lambda = 2, 2, 0, 0$
- $\vec{v}_1 = [1\ 1\ 1\ 1]$, $\vec{v}_2 = [1\ -1\ 1\ -1]$ are for $\lambda = 2$

🔍 For $\lambda = 0$: Solve $C\vec{x} = 0$

We solve:

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

That gives:

$$x_1 + x_3 = 0 \quad \text{and} \quad x_2 + x_4 = 0 \Rightarrow x_3 = -x_1,\ x_4 = -x_2$$

So general solution:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ -x_1 \\ -x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}$$

These are eigenvectors for $\lambda = 0$:

- $\vec{v}_3 = [1\ 0\ -1\ 0]$
- $\vec{v}_4 = [0\ 1\ 0\ -1]$

✅ Final Summary:

Matrix	Eigenvalue	Eigenvectors
A	4	$[1\ 1\ 1\ 1]^T$
	0	$[1\ -1\ 0\ 0]^T, [0\ 1\ -1\ 0]^T, [0\ 0\ 1\ -1]^T$
C	2	$[1\ 1\ 1\ 1]^T, [1\ -1\ 1\ -1]^T$
	0	$[1\ 0\ -1\ 0]^T, [0\ 1\ 0\ -1]^T$

Ask anything

28

20:41 Mon, 7 Apr 82%
ChatGPT

🔷 Step 1: Eigenvalues of B

We already know from graph theory and linear algebra that for a complete graph K_n :

- The adjacency matrix has:
  - One eigenvalue $\lambda = n - 1$ with multiplicity 1
  - One eigenvalue $\lambda = -1$ with multiplicity $n - 1$

So for $B = K_4$ adjacency matrix:

$$\boxed{\text{Eigenvalues of } B: \quad 3, -1, -1, -1}$$

🔷 Step 2: Determinant of B

Multiply the eigenvalues:

$$\det(B) = 3 \cdot (-1)^3 = \boxed{-3}$$

🔷 Step 3: Eigenvalues of $C = I - A$

Since $C = -B$, it's just the negative of matrix B. So the eigenvalues of C are the negatives of those of B:

$$\boxed{\text{Eigenvalues of } C: \quad -3, 1, 1, 1}$$

🔷 Step 4: Determinant of C

$$\det(C) = (-3) \cdot 1 \cdot 1 \cdot 1 = \boxed{-3}$$

+ Ask anything

Share a link to chat?
This creates a copy that others can chat with

$$\boxed{\text{Eigenvalues of } B: \quad 3, -1, -1, -1}$$

🔷 Step 2: Determinant of B

Multiply the eigenvalues:

$$\det(B) = 3 \cdot (-1)^3 = \boxed{-3}$$

🔷 Step 3: Eigenvalues of $C = I - A$

Since $C = -B$, it's just the negative of matrix B. So the eigenvalues of C are the negatives of those of B:

$$\boxed{\text{Eigenvalues of } C: \quad -3, 1, 1, 1}$$

🔷 Step 4: Determinant of C

$$\det(C) = (-3) \cdot 1 \cdot 1 \cdot 1 = \boxed{-3}$$

✅ Final Summary

Matrix	Eigenvalues	Determinant
$B = A - I$	$3, -1, -1, -1$	-3
$C = I - A$	$-3, 1, 1, 1$	-3

Let me know if you'd like the eigenvectors too or want to verify this step by step!

+ Ask anything

28 Subtract I from A in Problem 27. Find the λ's and then the determinants of

$$B = A - I = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \quad \text{and} \quad C = I - A = \begin{bmatrix} 0 & -1 & -1 & -1 \\ -1 & 0 & -1 & -1 \\ -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{bmatrix}.$$

29 (Review) Find the eigenvalues of A, B, and C:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 3 & 0 & 0 \end{bmatrix} \quad \text{and} \quad C = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix}.$$

30 When $a + b = c + d$ show that $(1, 1)$ is an eigenvector and find both eigenvalues:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

31 If we exchange rows 1 and 2 *and* columns 1 and 2, the eigenvalues don't change. Find eigenvectors of A and B for $\lambda = 11$. *Rank one gives* $\lambda_2 = \lambda_3 = 0$.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 6 & 3 \\ 4 & 8 & 4 \end{bmatrix} \quad \text{and} \quad B = PAP^{\mathrm{T}} = \begin{bmatrix} 6 & 3 & 3 \\ 2 & 1 & 1 \\ 8 & 4 & 4 \end{bmatrix}.$$

32 Suppose A has eigenvalues $0, 3, 5$ with independent eigenvectors $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$.

(a) Give a basis for the nullspace and a basis for the column space.

(b) Find a particular solution to $A\boldsymbol{x} = \boldsymbol{v} + \boldsymbol{w}$. Find all solutions.

(c) $A\boldsymbol{x} = \boldsymbol{u}$ has no solution. If it did then _____ would be in the column space.

Challenge Problems

33 Show that $\boldsymbol{u}$ is an eigenvector of the rank one 2×2 matrix $A = \boldsymbol{u}\boldsymbol{v}^{\mathrm{T}}$. Find both eigenvalues of A. Check that $\lambda_1 + \lambda_2$ agrees with the trace $u_1v_1 + u_2v_2$.

34 Find the eigenvalues of this permutation matrix P from $\det(P - \lambda I) = 0$. Which vectors are not changed by the permutation? They are eigenvectors for $\lambda = 1$. Can you find three more eigenvectors?

$$\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}$$

6.1

Final Summary:

Matrix	Eigenvalue	Eigenvectors
A	4	$[1\ 1\ 1\ 1]^T$
	0	$[1\ -1\ 0\ 0]^T, [0\ 1\ -1\ 0]^T, [0\ 0\ 1\ -1]^T$

20:59 Mon, 7 Apr • 80%
ChatGPT

Step 1: Eigenvector equation

We know the eigenvalue equation is:

$$A \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \lambda \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

Multiplying the matrix by the eigenvector:

$$A \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \begin{vmatrix} a+b \\ c+d \end{vmatrix}$$

For $\begin{bmatrix} 1 & 1 \end{bmatrix}$ to be an eigenvector, we require:

$$\begin{vmatrix} a+b \\ c+d \end{vmatrix} = \lambda \begin{vmatrix} 1 \\ 1 \end{vmatrix} = \begin{vmatrix} \lambda \\ \lambda \end{vmatrix}$$

This gives us the system:

$$a+b=\lambda \quad \text{and} \quad c+d=\lambda$$

Since $a+b=c+d$, we can conclude that $\lambda = a+b$ (which is also $\lambda = c+d$).

Thus, the first eigenvalue is:

$$\lambda_1 = a+b$$

Step 2: Find the second eigenvalue λ_2

The sum of the eigenvalues of a matrix is equal to its trace. The trace of matrix A is:

$$\text{trace}(A) = a+d$$

Since the sum of the eigenvalues is equal to the trace, we have:

$$\lambda_1 + \lambda_2 = \text{trace}(A) = a+d$$

From the previous step, we know that $\lambda_1 = a+b$, so we can solve for λ_2 :

$$\lambda_2 = \text{trace}(A) - \lambda_1 = (a+d) \quad - b) = d-b$$

Ask anything



21:13 Mon, 7 Apr • 79%
ChatGPT

(a) Give a basis for the nullspace and the column space

- Nullspace $\mathcal{N}(A) = \mathbf{x} : A\mathbf{x} = 0$

  Since $A\mathbf{u} = 0$, $\mathbf{u}$ spans the nullspace.

  ✅ Basis for the nullspace: $\mathbf{u}$

- Column space $\mathrm{Col}(A) = \mathrm{range}(A)$

  Since $A\mathbf{v} = 3\mathbf{v}$ and $A\mathbf{w} = 5\mathbf{w}$, these two vectors are in the image of A. They are linearly independent, so they form a basis.

  ✅ Basis for the column space: $\mathbf{v}, \mathbf{w}$

(b) Find a particular solution to $A\mathbf{z} = \mathbf{v} + \mathbf{w}$

Let's guess a particular solution as a linear combination:

$\mathbf{z}_p = a\mathbf{v} + b\mathbf{w}$

Apply A:

$A\mathbf{z}_p = aA\mathbf{v} + bA\mathbf{w} = a(3\mathbf{v}) + b(5\mathbf{w}) = 3a\mathbf{v} + 5b\mathbf{w}$

We want:

$A\mathbf{z}_p = \mathbf{v} + \mathbf{w} \Rightarrow 3a\mathbf{v} + 5b\mathbf{w} = \mathbf{v} + \mathbf{w}$

Match coefficients:

- $3a = 1 \Rightarrow a = \frac{1}{3}$
- $5b = 1 \Rightarrow b = \frac{1}{5}$

✅ Particular solution:

$\mathbf{z}_p = \frac{1}{3}\mathbf{v} + \frac{1}{5}\mathbf{w}$

✅ All solutions:

Ask anything

31

Row Reduction to Solve Linear System

Let's go through the row reduction process step-by-step to solve the system $(A - 11I)\vec{x} = 0$ where $A - 11I = \begin{bmatrix} -10 & 2 & 1 \\ 3 & -5 & 3 \\ 4 & 8 & -7 \end{bmatrix}$.

Step 1: Get leading 1 in row 1
Divide Row 1 by -10:

$$R_1 \leftarrow \frac{1}{-10} R_1 \Rightarrow \begin{bmatrix} 1 & -\frac{1}{5} & -\frac{1}{10} \\ 3 & -5 & 3 \\ 4 & 8 & -7 \end{bmatrix}$$

Step 2: Eliminate below pivot in column 1
Subtract 3 times Row 1 from Row 2 and 4 times Row 1 from Row 3:

$$R_2 \leftarrow R_2 - 3R_1$$
$$R_3 \leftarrow R_3 - 4R_1$$

Resulting matrix:

$$\begin{bmatrix} 1 & -\frac{1}{5} & -\frac{1}{10} \\ 0 & -\frac{22}{5} & \frac{33}{10} \\ 0 & \frac{44}{5} & -\frac{33}{5} \end{bmatrix}$$

Step 3: Make pivot in Row 2 a 1
Multiply Row 2 by $-\frac{5}{22}$:

$$R_2 \leftarrow -\frac{5}{22} R_2 \Rightarrow \begin{bmatrix} 1 & -\frac{1}{5} & -\frac{1}{10} \\ 0 & 1 & -\frac{3}{4} \\ 0 & \frac{44}{5} & -\frac{33}{5} \end{bmatrix}$$

Step 4: Eliminate above and below pivot in column 2
Add $\frac{1}{5}$ times Row 2 to Row 1 and subtract $\frac{44}{5}$ times Row 2 from Row 3:

$$R_1 \leftarrow R_1 + \frac{1}{5} R_2$$
$$R_3 \leftarrow R_3 - \frac{44}{5} R_2$$

Resulting matrix:

$$\begin{bmatrix} 1 & 0 & -\frac{1}{4} \\ 0 & 1 & -\frac{3}{4} \\ 0 & 0 & 0 \end{bmatrix}$$

Step 5: Back-substitute

The system of equations is:

$$x_1 - \frac{1}{4}x_3 = 0 \Rightarrow x_1 = \frac{1}{4}x_3$$
$$x_2 - \frac{3}{4}x_3 = 0 \Rightarrow x_2 = \frac{3}{4}x_3$$
$$x_3 = x_3$$

Let $x_3 = 4t$, then $x_1 = t$, $x_2 = 3t$, and $x_3 = 4t$. The solution is:

$$\vec{x} = t \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

Final Answer

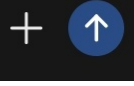
The solution to the system is all scalar multiples of the vector $\begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$.



New chat

Message DeepSeek

DeepThink (R1) Search



32 → before 31